Lakshmi Precision Technologies Limited

(Formerly Known as Lakshmi Precision Tools Limited)

About US

Lakshmi Precision Technologies Limited (LPT), (Formerly Known as Lakshmi Precision Tools Limited) a public limited company, established in 1983 where it is chosen to focus on Metal Parts, sub-assemblies for Textile, Engineering & Medical applications.

Our new division "**LPT – Tech serve**" is focus on **Global Engineering Solutions and Services**

LPT specializes in production of Thin-Walled High Pressure / Gravity Die Castings, High Precision Machined Components, Fabricated **Sheet Metal Components** & Ready-to-fit Sub-assemblies for Industrial applications. Our infrastructure includes a full-fledged Tool Room, R&D Centre, **Chemical & Mechanical Testing Laboratories** and latest version of Design Software to enhance designing abilities.

LPT is an ISO 9001: 2015 / ISO 14001:2015, ISO 45001:2018 & 5S Certified Facility

LPT- Philosophy

H&S
HEALTH & SAFETY

- ✓ Ensure Safe Work Envronment
- ✓ Zeor Non-Compliance

Q
QUALITY

- ✓ Continuous Improvement
- ✓ Multi Skilling

C
COST

- ✓ Productivity
- ✓ Value Engineering

D
DELIVERY

- ✓ On - time Delivery
- ✓ Every-time and All Time

D
DEVELOPMENT

- ✓ Technology Advancement
- ✓ First Time Right

Our Values

MISSION

Create Superior Value For the Stake Holders through Our Products & Services to ensure Sustainable Growth and Operational Excellence using Innovative Methods, Process Driven Approaches and Eco- friendly Solution

VISION

To be a Premier Multi-Industry Company Known for Manufacturing Excellence through Ethical Business Practices and Quality Standards.

VALUES

Accountability: For our words, our actions, and ourresults.

Respect: Everyone and treat people with nobility and professionalism.

Integrity: Trust through responsible actions and honest relationships.

Teamwork: Achieve more when we are collective and all work together

Our- Corporate Companies & Associate Companies

LPT

Corporate

LECS

LEDL

LMW

(Textile Machinery Division)

Lakshmi LRT

Foundry Division
(Steel Castings)

LMW

Machine Tool Division
(Turning & Milling Centre)

LMW

Foundry Division
(SG Iron Castings)

Our- Associate Companies

Engineering & Manufacturing

(Tooling)

(Injection Moulding)

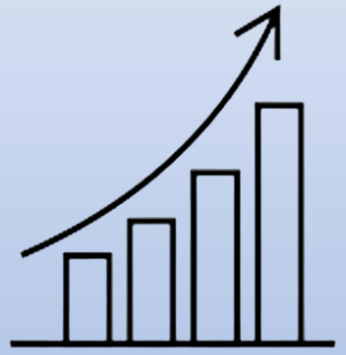
Advance Technology Centre

(Aerospace & Composite)

LPT – IN Brief

Genesis : EUR 0.6 Bn + from 60 Years old Group in Coimbatore, India. (Manchester of South India)

Excellent Operational & Financial Track record



- Annual Turnover of EUR 67 Mn and growing rapidly
- Industries: Textile, Machine tool , Wind & Solar Energy, EV, Petrochemical, Railways, Agriculture, Healthcare, Retail, Auto, General Engineering, Switchgear, Railways & Tech serve

- ISO-9001-2015, ISO45001-2018, EHS ISO-14001-2015, IATF-16949 Certified Manufacturing Facility
- Products CE Marked, UL Compliant (selectively)
- SAP ERP Full system capability

Industry Verticals

Automotives	01	06	Crushing & Mining Products
Energy	02	07	Food Process & Dairy Machines
Embedded & Software Solution	03	08	Textile Machinery
Agriculture Equipment's	04	09	Medical & Healthcare
Trucks & Off-Highway	05	10	Aero Space

Why LPT-Techserve?

Trusted engineering partner for global OEMs on their NPD and localization programs. Increased market share with Accelerated product development.

End to End Solution

Expertise on complete engineering ownership, from concept design to manufacturing support and aftermarket solution.

Vast technical and process knowledge acquired

Established centre-of-Excellence (CoE) to encourage innovation and deliver value to our clients.

To provide superior control over outcome and costs and risks bearing by LPT-techserve Highly Skilled and Scalable talent Pool.

Global benchmark process have been deployed to ensure the highest levels of quality and productivity, technology and process.

Global Support

Support across geographical regions throughout office or Engineering Development Centre's.

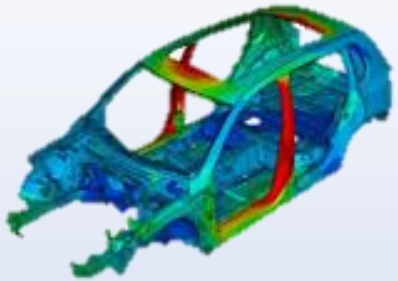
LPT-Techserve – Offerings – Engineering Service

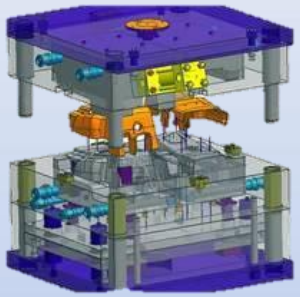
- Concept Development
- Styling Feedback
- Functional design review
- Design Development
- Electrical design
- Harness development

Product Design

Design Validation & Virtual Simulation

- FE Molding
- Crash & Safety
- Strength & Durability
- CFD
- Mfg. Process Simulation (DIS)
- Supply chain



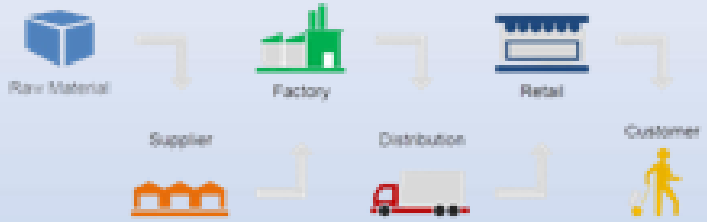


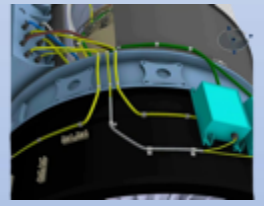
- Process Planning
- Digital Manufacturing
- Workstation Design
- Tools and Equipment Design
- Packaging Design
- Computer Aided Manufacturing
- Bench Marking

Manufacturing Engineering & Solutions

Reverse Engineering & VAVE

- Ideation
- Functional Analysis
- Evaluation
- Selection





Electrical Design & Development

Prototyping

- 3D Scanning
- Data Cleansing
- 3D Model Generation

- Process Automation
- Toolkit development
- Design Automation

Automation

Data Migration & Tech. Publication

- 2D to 3D Creation
- CAD Platform Migration
- Technical Documentation and User manuals

LPT-Techserve – Offerings – Software Service

- SAP TAO, OOABAP, SAP CRM, SAP APO,
- Enterprise Mobile Consulting (MAM, MDM) on turnkey basis,
- Salesforce

ERP & CRM

BI/Data Warehousing /BPM

- Talend, MSBI, Data stage - Data Quality, Data Integration and Data Governance - BigData, MetaData, DataScientist, AB Initio,
- Micro strategy, SAP BI/ BW, AGILE, Qlick view, Lombardi

Application Development

IT Services

- Infrastructure
- Training
- Testing

Out System

- Appian,
- Microsoft PowerApps
- Mendix

Portal & Content Management

- IBM Websphere Portal
- IBM Weblogic Portal
- Vignette
- Documentum

PLM

- CAD Integration
- Solution Architect
- Team Centre Developing

IOT

UI, UX, Embedded, Edge

LPT-Techserve – Offerings – Talent Solutions

Talent Solutions specializes in providing end to end recruiting solutions and consulting.

Our services are:

- Permanent Recruitment
- Contractor Recruitment
- Recruitment Process Outsourcing
- Short term Project Outsourcing
- Day one deployable resource - Trained Fresher's

- ✓ 25 Dedicated Recruiters for Indian Market
- ✓ 100+ Closures per QTR
- ✓ 237+ Contract resources deployed – Pan India
- ✓ Lead time
  - ✓ 8 hours for Vanilla skills
  - ✓ 24 hours for Nish Skills
  - ✓ 48 hours for Super Nish Skills

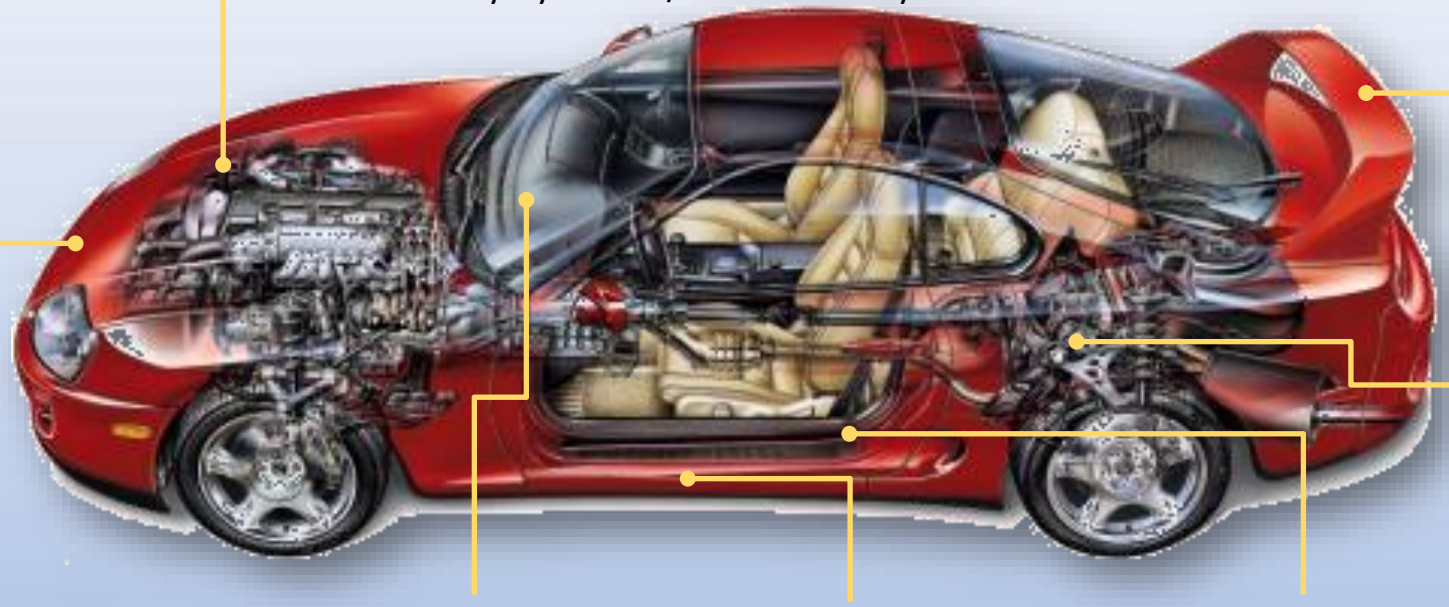
Our Capabilities

Automotive Areas of Expertise

Power Train
- Engine Castings
- Engine Testing & Performance
- Engine Integration
- Motor Design

Digital Validation
- Static / Dynamic / Crash / Fatigue / Impact
- Modal /Buckling / CFD / Optimization
- Multi-Body Dynamics / Thermal analysis

Exteriors
- Bumpers
- Exterior Trim
- Lighting systems
- Front-ends



Electricals
- Wire Harness
- Electrical Distribution Modules

Closures
- Doors
- Hoods
- Deck Lids
- Trunk lids
- Mirrors

Electronics
- Engine Management Systems
- Telematic systems
- Safety Systems
- Comfort systems

Interiors
- Seating Systems
- IP & Consoles
- OVHS
- Door Trims
- Safety Systems
- Body-side Trims
- Trunk Systems
- HVAC

B I W
- Under body
- Floor, Sills
- Firewalls, Roof
- Side Panels

Underbody
- Chassis
- Suspensions
- Braking Systems
- Wheel /Axles
- Sub frames

Picture Courtesy: Car Illustration.com

Expertise: Body

Mechanical Design

- Cockpit Module
- Seating
- Safety Systems

Embedded Design

- Instrument Cluster
- Infotainment
- Telematics

CAD

- Class A surfacing
- Instrument panel design
- Seating Structures & Foam Design
- Airbag & Retractor Design
- Interior Trims

CAE

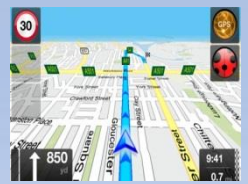
- Mold flow Analysis
- Airbag Simulation
- Retractor Analysis
- Seat Indent Tests
- FEA & CFD

Hardware

- Schematic design
- CAD Library Development
- SI and PI Analysis
- Timing analysis & signal integrity analysis

Software

- Instrument cluster development
- Genivi Based Infotainment & Telematics platform Development
- HMI Development based on *(EBguide, Altia, QT, HTML5)*
- Connectivity & Interoperability Testing *(Bluetooth/ WIFI/ Vehicle Network)*
- Platform Testing & Validation



Expertise: Interiors

Mechanical Design

- Body in White
- Exteriors
- Closures

Embedded Design

- Body Control Module
- Advanced Driver Assistance

CAD

- Upper body Design
  (Roof,Pillars,Panels,Door Frames)
- Underbody Design
  (Floor,Sills,Wheel House)
- Closure Design
  (Doors, Tailgates,Trunklid,Hood etc.)
- Exteriors Design
  (Front/Rear Fascia,Fenders,Light,Grilles)

CAE

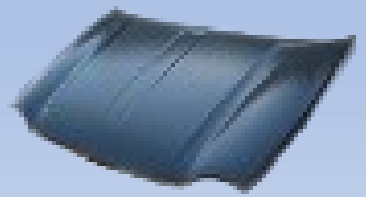
- Fatigue Analysis
- Durability Analysis
- Roof Crush
- Quasi-static stiffness
- Pedestrian Impact

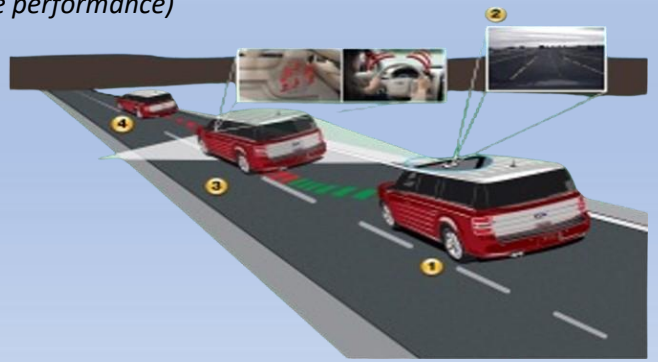
Hardware

- PCB Layout Design
- High power board design
- Risk Analysis
- Hazard Analysis

Software

- Body Control Module Development Compliant to AUTOSAR & ISO 26262
- Sensor & Vision based Image processing algorithm development
  (Lane Departure, Collision Avoidance, Drowsiness monitoring etc)
- Airbag Control Unit Design
  (Linux porting & Boot-up optimization for real time performance)





Expertise: Powertrain

Mechanical Design

- Engine
- Transmission
- Exhaust

Embedded Design

- Engine Management System
- Battery Management System
- OBD

CAD

- Engine Design (Castings,Mounting,Layout studies)
- Transmission Design
- (Gears, Shafts & Std Parts design)
- Exhaust System design
- (Manifolds, Tail pipes, Filters etc.)
- Tear Down & Benchmarking

CAE

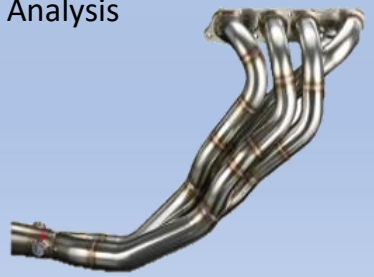
- Combustion Simulation
- Fatigue Analysis
- Exhaust Simulation
- CFD Air Intake System
- CHT Analysis

Hardware

- Schematics Design & Circuit Simulation
- HW Driver Development
- EMI/EMC testing as per SAE J 1113 and 1455 standards

Software

- AUTOSAR Compliant ECU Software Development
- Model Based Feature Development
- Diagnostics Software Development
- (CAN,LIN,FLEXRAY etc)
- HIL Testing
- (Static Simulator – Pi AutoSim)



AUTOSAR

Expertise: Steering & Suspension

Mechanical Design

- Chassis
- Steering
- Brakes & Suspension

Embedded Design

- Electronic power Steering
- ESP
- ABS

CAD

- Steering System Design
- Suspension Design
- (Struts,springs,Modules etc)
- Front/Rear Axle Design
- Brake System Development
- GD&T Support

CAE

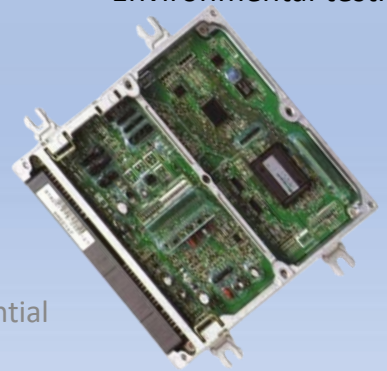
- Coupled Analysis of Brake Drum
- Vibration
- MBD Analysis
- Rigidity Analysis

Hardware

- High Voltage & High current Design
- Noise analysis
- Interface circuit design
- Environmental testing

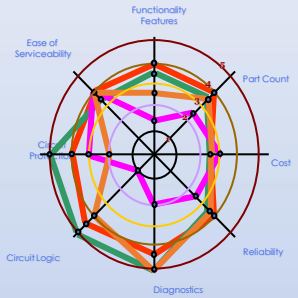
Software

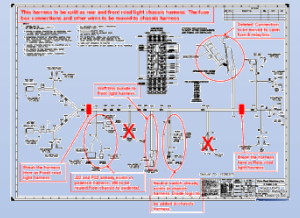
- Model Based Design of Electronic power steering based on ISO 26262 requirement
- Torque Control Algorithm Development
- Software Testing & EOL Testing
- Vehicle Level Integration with Active & Passive Safety Systems

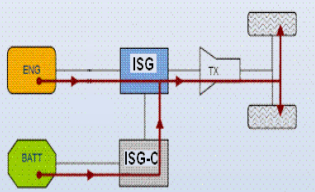


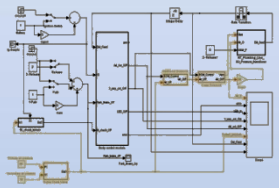
Capabilities in Electrical Design Services:

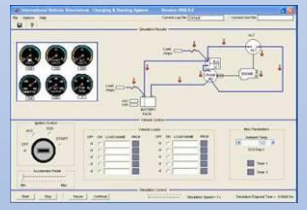
Requirement Capture & Study → Proof of Concept & Simulation → Component Development* → Hardware Design → Verification & Validation → Service & Mfg. Support

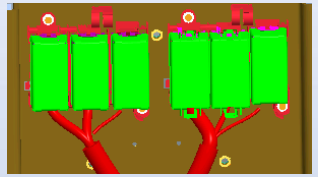


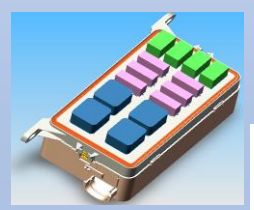


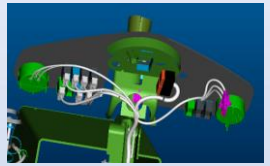


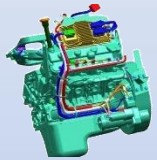


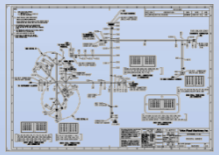


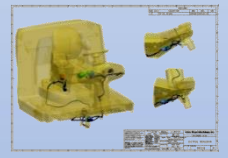


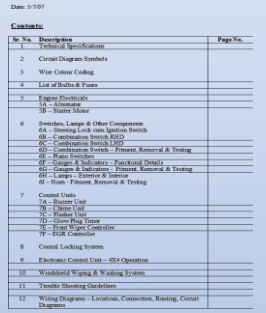












Date: 5/7/07

Contents:

Sr. No.	Description	Page No.
1	Technical Specifications	
2	Circuit Diagram Symbols	
3	Wire Colour Coding	
4	List of Bulbs & Fuses	
5	Engine Electricals 5A – Alternator 5B – Starter Motor	
6	Switches, Lamps & Other Components 6A – Steering Lock cum Ignition Switch 6B – Combination Switch RHD 6C – Combination Switch LHD 6D – Combination Switch – Fitment, Removal & Testing 6E – Piano Switches 6F – Gauges & Indicators – Functional Details 6G – Gauges & Indicators – Fitment, Removal & Testing 6H – Lamps – Exterior & Interior 6I – Horn – Fitment, Removal & Testing	
7	Control Units 7A – Buzzer Unit 7B – Chime Unit 7C – Flasher Unit 7D – Glow Plug Timer 7E – Front Wiper Controller 7F – EGR Controller	
8	Central Locking System	
9	Electronic Control Unit – 4X4 Operation	
10	Windshield Wiping & Washing System	
11	Trouble Shooting Guidelines	
12	Wiring Diagrams – Location, Connection, Routing, Circuit Diagrams	

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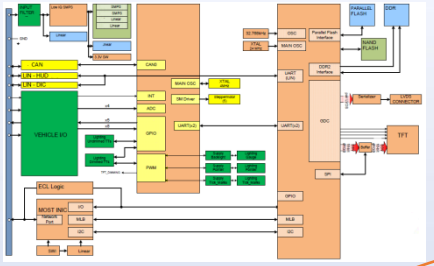
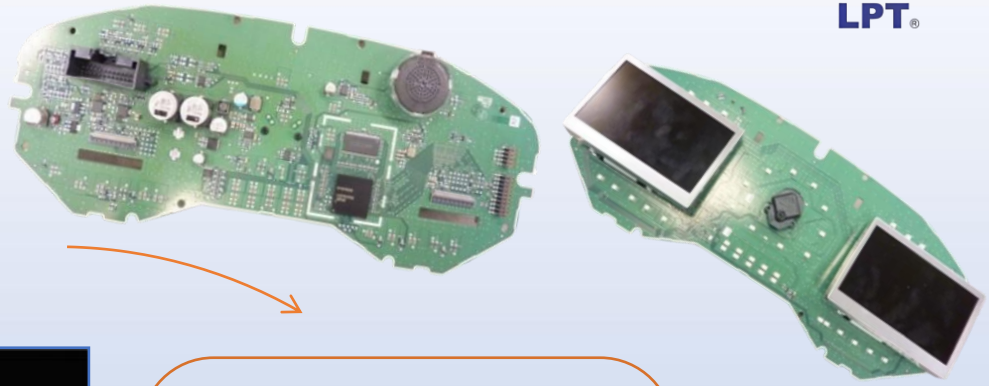
Embedded Case Studies

Instrument Cluster Capabilities



Requirement Gathering

Comb through OEM specs to segregate and ID requirements



Testing and Warranty

ECU Bench Level Testing

Verification and Validation

EMC compliance support

Warranty return analysis

ECU Design

GPU with DDRx memory

32-Bit MCU with steppers

Power supply (IOD requirements)

MOST, CAN, LIN Interface

Analog and Switch Interfaces

SW/HMI Development

Architecture, Algorithm, Boot loader

Protocol stack, Diagnostics, Integration

HMI – Function, Behavior and Style

HMI Expertise

Capabilities

- Expertise in HMI development for Automotive TIER 1 companies.
- Software development platform for 3D graphical human machine interfaces (HMI/GUI) on automotive instrument clusters and infotainment systems.
- HMI capabilities includes creation of 2D & 3D widgets, 2D & 3D graphics rendering via multiple layers.
- HMI capabilities includes Static and Dynamic animation management, resolution and positioning.
- Template Development
- Template Development in Table list template, Background template, Button Template with all required features.

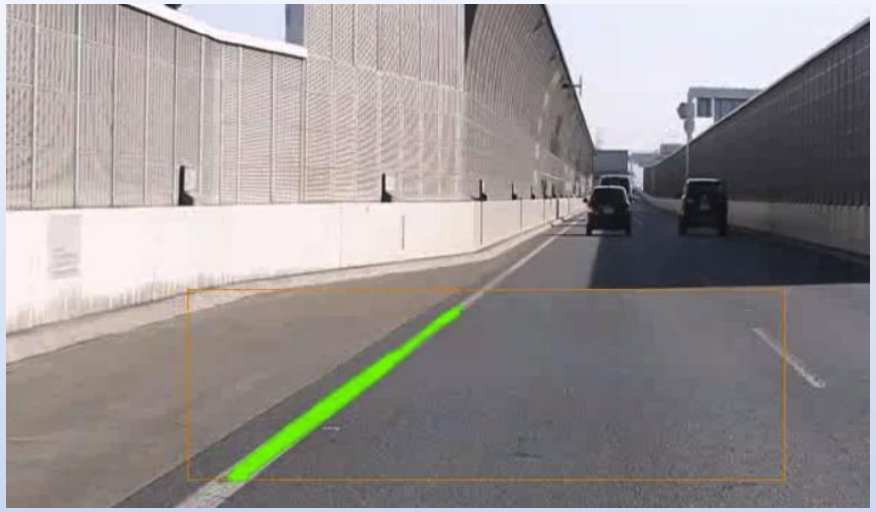
Service offerings

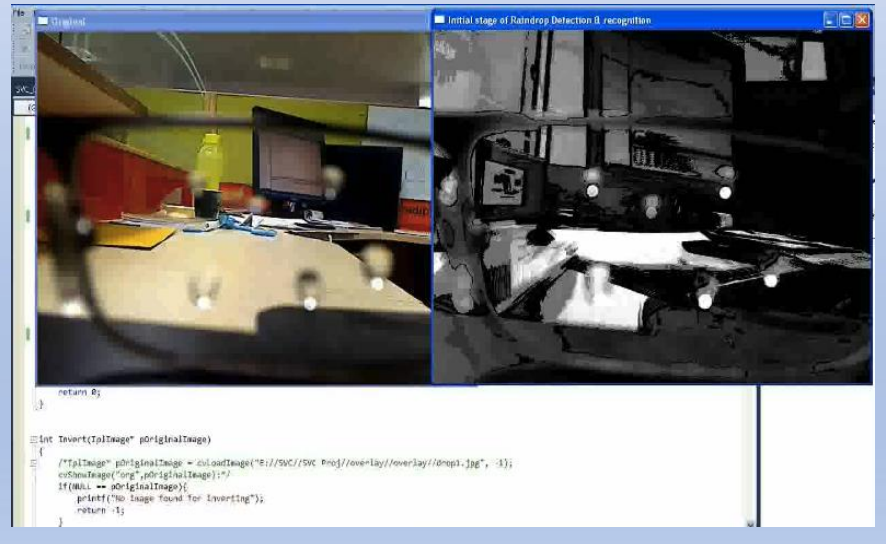
- Supported HMI development tools are Elektrobit Guide Studio with version 2.x & 5.x, Altia, Edamama, Fujitsu CGI Studio, PEG,QT
- Supported programming languages are C, C++, C#.Net and JAVA.
- Supported operating systems are Linux, Windows, wince, eCos.
- Device drivers / BSP for peripherals like Touch Screen XVGA, UART, USB Host, DAC, NVRAM, Flash, I2C, Speaker.

Highlights

- HMI Screens with frame rate of 30fps and more with graphic resolution of 1280 x 1024 pixels.
- Defining dynamic system behavior for interaction between HMI and application.
- Modeling of menu structure using states with UML compliant format.
- Rapid prototyping on desktop PC environment.
- Multilanguage supports.
- Modularity to configure screen data using XML file.

Other Project Case Studies





Our Customers (USA Based)

Ingersoll Rand

APPLIED MATERIALS make possible

AKG

EATON

FIAT
Maserati
RAM
CHRYSLER
DODGE
Jeep
ALFA ROMEO
STELLANTIS
CITROËN
LANCIA
DS AUTOMOBILES
PEUGEOT
VAUXHALL

wipro
GE Healthcare

KONAMI

Our Customers

INTEVA PRODUCTS

Delphi-TVS Technologies

IFB Automotive Private Limited

KONGOVI

SONA COMSTAR

SPARK MINDA VAST
MINDA VAST ACCESS SYSTEMS

APTIV

MAHLE
Driven by performance

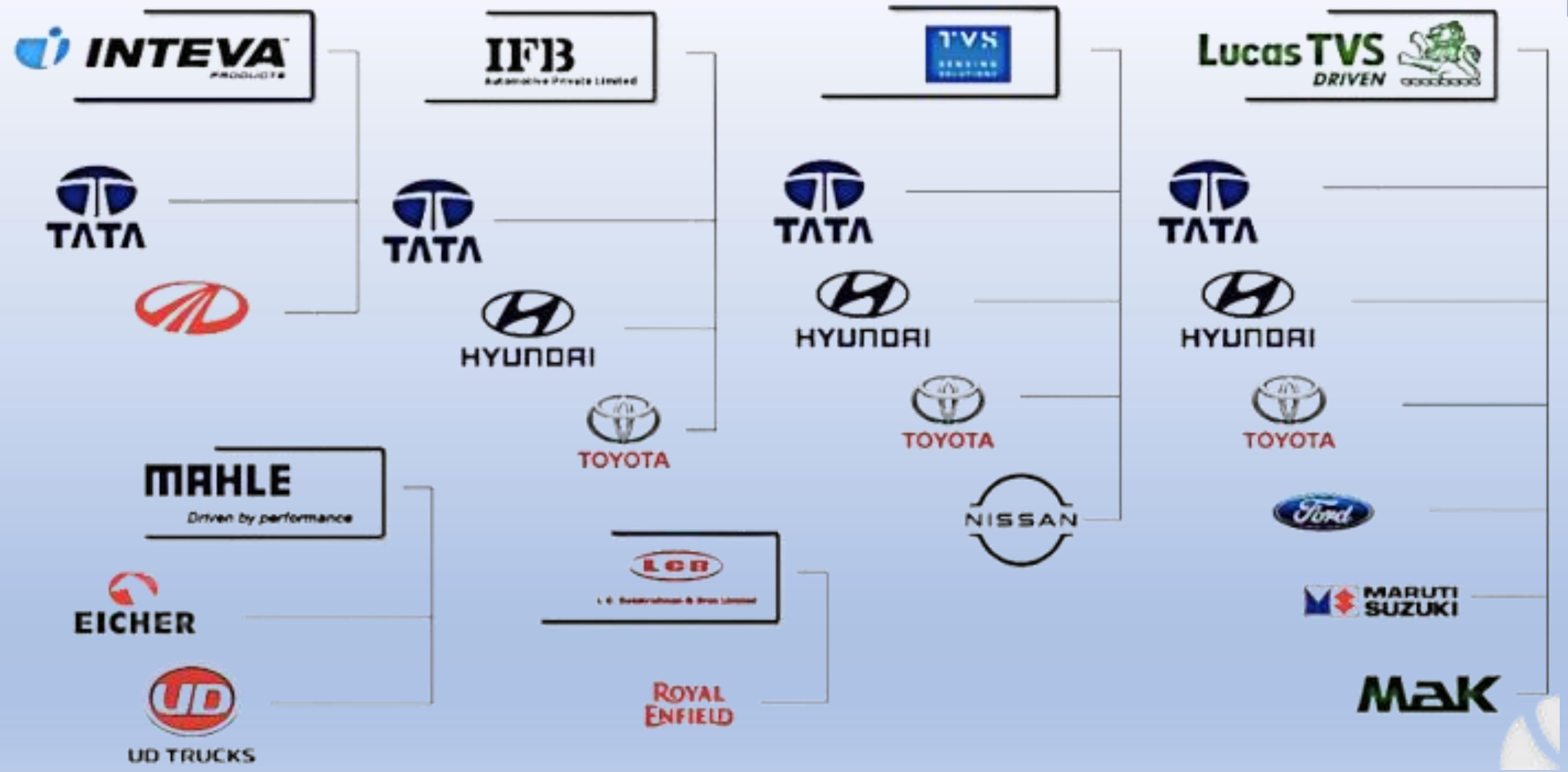
Lucas TVS DRIVEN

ITW

GRINNTECH

SIMPLE

Our Customers



Our Customers

GE Energy

auma
Solutions for a world in motion

Lucas-TVS Limited

LMW

Rossi
Hulsrelt Group

GILBARCO VEEDER-ROOT

GE Industrial Systems

MACHINE TOOL DIVISION

KSB

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MAKINO

ELGi
Always Better.

vortex
Helping banks reach out

Fowler westrup

ReGen POWERTECH

TMEIC
We drive industry

GE Healthcare

Rane TRW

SIEMENS Gamesa
RENEWABLE ENERGY

murugappa

SCHWING Stetter

ZAHORANSKY
PERFECTLY DIFFERENT

Shanthi Gears

LIEBHERR

TOSHIBA MACHINE

Bonfiglioli

ElectroMech

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